

Assignment0: Hadoop Map reduce

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Section: C

1. Verify the number of records:

```
~/assign0 PES2UG24CS157  
> vim sales.json  
~/assign0 PES2UG24CS157  
> vim convert.py  
~/assign0 PES2UG24CS157  
> python3 convert.py  
~/assign0 PES2UG24CS157  
> ls  
convert.py  sales.json  sales.jsonl  
~/assign0 PES2UG24CS157  
> wc -l sales.jsonl  
15 sales.jsonl  
~/assign0 PES2UG24CS157  
> |
```

2. `cat sales.jsonl`

```
~/assign0 PES2UG24CS157
>> cat sales.jsonl
{"order_id": 1001, "product": "Laptop", "category": "Electronics", "quantity": 1, "price": 55000}
{"order_id": 1002, "product": "Mouse", "category": "Electronics", "quantity": 2, "price": 750}
{"order_id": 1003, "product": "Keyboard", "category": "Electronics", "quantity": 1, "price": 1150}
{"order_id": 1004, "product": "T-Shirt", "category": "Clothing", "quantity": 3, "price": 800}
{"order_id": 1005, "product": "Jeans", "category": "Clothing", "quantity": 2, "price": 1400}
{"order_id": 1006, "product": "Headphones", "category": "Electronics", "quantity": 2, "price": 2000}
{"order_id": 1007, "product": "Shoes", "category": "Footwear", "quantity": 1, "price": 2500}
{"order_id": 1008, "product": "T-Shirt", "category": "Clothing", "quantity": 2, "price": 800}
{"order_id": 1009, "product": "Notebook", "category": "Stationery", "quantity": 5, "price": 100}
{"order_id": 1010, "product": "Pen", "category": "Stationery", "quantity": 10, "price": 20}
{"order_id": 1011, "product": "Backpack", "category": "Accessories", "quantity": 1, "price": 1800}
{"order_id": 1012, "product": "Shoes", "category": "Footwear", "quantity": 2, "price": 2100}
{"order_id": 1013, "product": "Monitor", "category": "Electronics", "quantity": 1, "price": 12000}
{"order_id": 1014, "product": "Jacket", "category": "Clothing", "quantity": 1, "price": 3000}
{"order_id": 1015, "product": "Watch", "category": "Accessories", "quantity": 2, "price": 2200}
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```

3. Pass the input dataset to the mapper:

```
cat sales.jsonl | ./mapper.py
```

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> cat sales.jsonl | ./mapper.py

Electronics	55000
Electronics	1500
Electronics	1150
Clothing	2400
Clothing	2800
Electronics	4000
Footwear	2500
Clothing	1600
Stationery	500
Stationery	200
Accessories	1800
Footwear	4200
Electronics	12000
Clothing	3000
Accessories	4400

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4. Sort the output from mapper using:

```
cat sales.jsonl | ./mapper.py | sort
```



```
~/assign0 PES2UG24CS157  
> cat sales.jsonl | ./mapper.py | sort  
Accessories      1800  
Accessories      4400  
Clothing         1600  
Clothing         2400  
Clothing         2800  
Clothing         3000  
Electronics      1150  
Electronics      12000  
Electronics      1500  
Electronics      4000  
Electronics      55000  
Footwear         2500  
Footwear         4200  
Stationery       200  
Stationery       500
```

5. sorted output provided as input to the Reducer:

```
cat sales.jsonl | ./mapper.py | sort | ./reducer.py
```

```
~/assign0 PES2UG24CS157 ▶
▶ cat sales.jsonl | ./mapper.py | sort | ./reducer.py
Accessories      6200.00
Clothing         9800.00
Electronics      73650.00
Footwear         6700.00
Stationery       700.00
~/assign0 PES2UG24CS157 ▶
▶ |
```

6. Ingest the dataset into HDFS

```
~/assign0 PES2UG24CS157  
➤ start-dfs.sh  
start-yarn.sh  
Starting namenodes on [localhost]  
Starting datanodes  
Starting secondary namenodes [server1]  
Starting resourcemanager  
Starting nodemanagers  
~/assign0 PES2UG24CS157  
➤ jps  
3285 SecondaryNameNode  
3494 ResourceManager  
4009 Jps  
3834 NodeManager  
2893 NameNode  
~/assign0 PES2UG24CS157  
➤ |
```

```
~/assign0 PES2UG24CS157
>> hdfs dfs -mkdir -p /input
~/assign0 PES2UG24CS157
>> hdfs dfs -put sales.jsonl /input/
~/assign0 PES2UG24CS157
>> hdfs dfs -ls /input
Found 1 items
-rw-r--r-- 1 dharani supergroup 1437 2026-08-22 15:04 /input/sales.jsonl
~/assign0 PES2UG24CS157
>> hdfs dfs -cat /assignment0/input/sales.jsonl
cat: '/assignment0/input/sales.jsonl': No such file or directory
~/assign0 PES2UG24CS157
>> hdfs dfs -cat /input/sales.jsonl
{"order_id": 1001, "product": "Laptop", "category": "Electronics", "quantity": 1, "price": 55000}
{"order_id": 1002, "product": "Mouse", "category": "Electronics", "quantity": 2, "price": 750}
{"order_id": 1003, "product": "Keyboard", "category": "Electronics", "quantity": 1, "price": 1150}
{"order_id": 1004, "product": "T-Shirt", "category": "Clothing", "quantity": 3, "price": 800}
{"order_id": 1005, "product": "Jeans", "category": "Clothing", "quantity": 2, "price": 1400}
{"order_id": 1006, "product": "Headphones", "category": "Electronics", "quantity": 2, "price": 2000}
{"order_id": 1007, "product": "Shoes", "category": "Footwear", "quantity": 1, "price": 2500}
{"order_id": 1008, "product": "T-Shirt", "category": "Clothing", "quantity": 2, "price": 800}
{"order_id": 1009, "product": "Notebook", "category": "Stationery", "quantity": 5, "price": 100}
{"order_id": 1010, "product": "Pen", "category": "Stationery", "quantity": 10, "price": 20}
{"order_id": 1011, "product": "Backpack", "category": "Accessories", "quantity": 1, "price": 1800}
{"order_id": 1012, "product": "Shoes", "category": "Footwear", "quantity": 2, "price": 2100}
{"order_id": 1013, "product": "Monitor", "category": "Electronics", "quantity": 1, "price": 12000}
{"order_id": 1014, "product": "Jacket", "category": "Clothing", "quantity": 1, "price": 3000}
{"order_id": 1015, "product": "Watch", "category": "Accessories", "quantity": 2, "price": 2200}
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```


7. Run the MapReduce job on Hadoop

```
Starting nodemanagers
~/assign0 PES2UG24CS157
>> hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
    -files mapper.py, reducer.py \
    -input /input/sales.jsonl \
    -output /output \
    -mapper "python3 mapper.py" \
    -reducer "python3 reducer.py"
packageJobJar: [/tmp/hadoop-unjar12684584807212306832/] [] /tmp/streamjob18112653714752910892.jar tmpDir=null
2026-08-22 15:10:42,822 INFO client.DefaultNoHARMFaloverProxyProvider: Connecting to ResourceManager at loca
lhost/127.0.0.1:8032
2026-08-22 15:10:42,935 INFO client.DefaultNoHARMFaloverProxyProvider: Connecting to ResourceManager at loca
lhost/127.0.0.1:8032
2026-08-22 15:10:43,150 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-ya
rn/staging/dharani/.staging/job_1787411414951_0001
2026-08-22 15:10:43,904 INFO mapred.FileInputFormat: Total input files to process : 1
2026-08-22 15:10:44,833 INFO mapreduce.JobSubmitter: number of splits:2
2026-08-22 15:10:45,011 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1787411414951_0001
2026-08-22 15:10:45,011 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-08-22 15:10:45,167 INFO conf.Configuration: resource-types.xml not found
2026-08-22 15:10:45,168 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-08-22 15:10:45,524 INFO impl.YarnClientImpl: Submitted application application_1787411414951_0001
2026-08-22 15:10:45,565 INFO mapreduce.Job: The url to track the job: http://server1:8088/proxy/application_1
787411414951_0001/
2026-08-22 15:10:45,566 INFO mapreduce.Job: Running job: job_1787411414951_0001
2026-08-22 15:10:50,654 INFO mapreduce.Job: Job job_1787411414951_0001 running in uber mode : false
2026-08-22 15:10:50,655 INFO mapreduce.Job: map 0% reduce 0%
2026-08-22 15:10:56,769 INFO mapreduce.Job: map 100% reduce 0%
2026-08-22 15:11:00,811 INFO mapreduce.Job: map 100% reduce 100%
2026-08-22 15:11:02,838 INFO mapreduce.Job: Job job_1787411414951_0001 completed successfully
2026-08-22 15:11:02,916 INFO mapreduce.Job: Counters: 54
    File System Counters
        FILE: Number of bytes read=271
        FILE: Number of bytes written=840985
        FILE: Number of read operations=0
```

Check the output directory:

```
~/assign0 PES2UG24CS157
>> hdfs dfs -ls /output
Found 2 items
-rw-r--r-- 1 dharani supergroup 0 2026-08-22 15:10 /output/_SUCCESS
-rw-r--r-- 1 dharani supergroup 93 2026-08-22 15:10 /output/part-00000
~/assign0 PES2UG24CS157
>> |
```


Verify its content

```
~/assign0 PES2UG24CS157
>> hdfs dfs -cat /output/part-00000
Accessories      6200.00
Clothing         9800.00
Electronics      73650.00
Footwear         6700.00
Stationery       700.00
~/assign0 PES2UG24CS157
>> |
```

8. Verify the job through yarn

```
~/assign0 PES2UG24CS157
>> yarn application -list -appStates FINISHED
2026-08-22 15:14:49,121 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at localhost/127.0.0.1:8032
Total number of applications (application-types: [], states: [FINISHED] and tags: []):1

```

User	Application-Id	Application-Name	Application-Type
Progress	Queue	State	Final-State
		Tracking-URL	
application	1787411414951 0001	streamjob18112653714752910892.jar	
MAPREDUCE	dharani	default	FINISHED
SUCCEEDED		100% http://server1:19888/jobhistory/job/job_1787411414951 0001	

```
~/assign0 PES2UG24CS157
>> |
```

1: All Applications



Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	
1	0	0	1	0	<memory:0 B, vCores:1>

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes
1	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation
Capacity Scheduler	[memory-mb (unit=Mi), vcores]	<memory:1024, vCores:1>

Show 20 ▼ entries

ID	User	Name	Application Type	Application Tags	Queue	Application Priority	StartTime	LaunchTime
application_1787411414951_0001	dharani	streamjob18112653714752910892.jar	MAPREDUCE		default	0	Sat Aug 22 15:10:45 +0000 2026	Sat Aug 22 15:10:45 +0000 2026

Showing 1 to 1 of 1 entries